

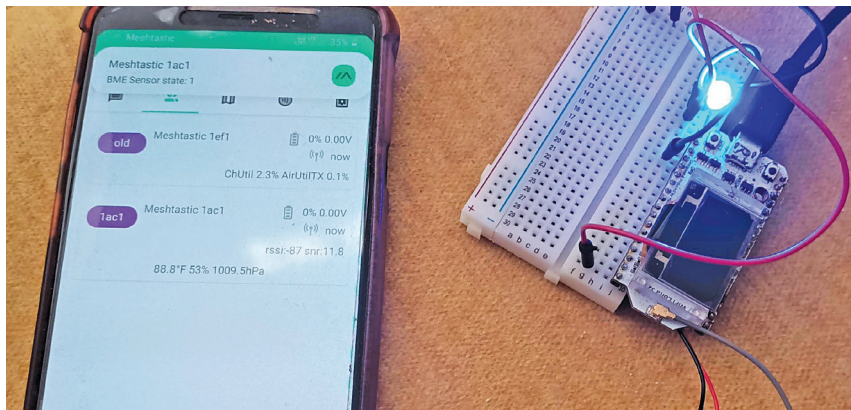


Fig. 9: LED alarm fired up

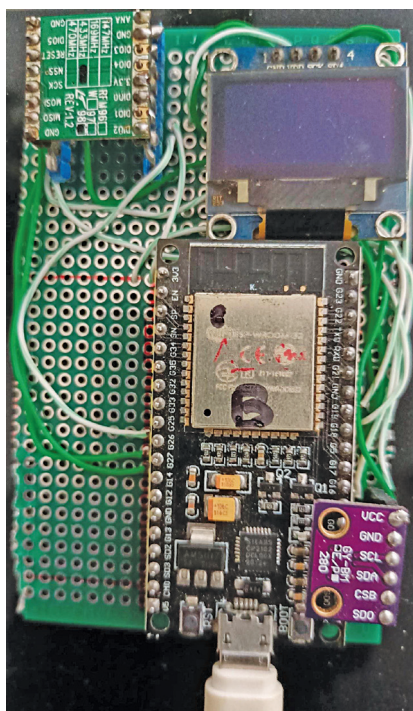


Fig. 10: Mesh node device with sensor

be visible against that node in the app-window. The app has the facility to attach a separate GPS sensor, for which you must manually set the GPIO pins for Tx and Rx: Radio configuration→Position. (Refer Fig. 8, where location is coming from the phone location service.) However, after setting the GPS sensor, the location data (latitude and longitude) would come from GPS sensor.

As per the site, only uBLOX, GLONASS GPS sensors are support-

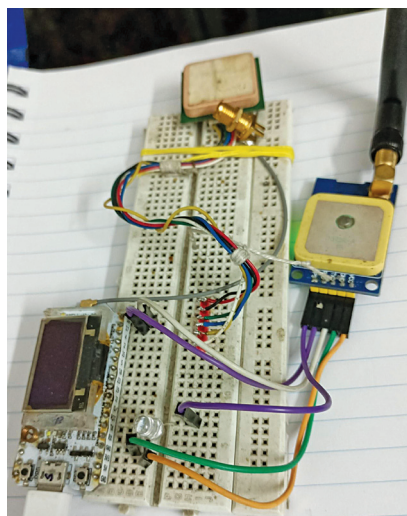


Fig. 11: Final prototype on bread board

ed at this moment. Author's old GPS sensor (V.KEL TTL GPS sensor) also worked when connected.

Network

The network can be switched on by clicking Radio Configuration→Network→Enable Wi-Fi→enter SSID and PSK. But doing this will make your node inaccessible by Bluetooth. Then you must access it by USB cable or network protocol using web server, which is not very friendly.

Doing a reversal of this is very frustrating. You have to reinstall the firmware again!

Typical uses

In a forest area, LoRa mesh can be used to monitor forest fires. In such cases, each node will monitor air temperature, velocity, and humidity of its zone. In an industrial area, such mesh can be used very effectively for monitoring fire hazards. In a city, by turning vehicles to LoRa nodes, the vehicles' speeds and positions can be monitored. In a township /housing complex, using V53LOX laser sensors, all the water tank levels can be monitored. It is possible to chat with the nodes and, using MQTT, it is even possible to reach other cities through Meshtastic LoRa networking.

Radio configuration→Module configuration→Store & Forward command works for ESP32 only. Here you can store a few records in the event of disconnection from node. However, Meshtastic does not recommend its use, normally.

Aftermath

The Meshtastic software has shown how a LoRa mesh can deploy the LoRa radios to get the best of networking for all good things like low power, great ranges, and impregnability. It's fun to play with the network. After setting up the nodes, one can disconnect the Android/iOS devices, and the nodes will continue to work automatically.

Fig. 9 shows how an LED alarm fires up. Fig. 10 shows a mesh node device along with its sensor. The final prototype that includes ESP32, LoRa, OLED screen, and BME 280 sensor is shown in Fig. 11. **EFY**

EFY Note

The source code of this project is available

for free download at
www.electronicshobby.com



Somnath Bera is an electronics enthusiast. He is a freelancer and has contributed several articles across the globe